📋 PROJECT REPORT

LoanLess AI

Intelligent Financial Analysis Platform — An AI-powered BFSI web platform for loan eligibility analysis, EMI calculation, credit score evaluation, and AI-powered financial guidance.

HTML5 | CSS3 | JavaScript | Claude AI | Google Apps Script | Google Sheets

Report Date: October 26, 2026

Platform Type: Static Frontend (MPA)

Total Lines of Code: ~3,150+

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1 Executive Summary

LoanLess AI is a fully client-side, zero-dependency web platform built with HTML5, CSS3, and Vanilla JavaScript. It provides four core financial tools — Loan Eligibility Checker, Credit Score Analyzer, EMI Calculator, and an AI Financial Advisor — all powered by Claude AI with intelligent local fallback responses.

The platform uses a serverless Google Apps Script + Google Sheets backend for data persistence and features a premium glassmorphism dark-theme design system with 50+ CSS custom properties. It targets the Indian BFSI market with INR formatting, CIBIL credit scoring (300–900), and Indian lending eligibility rules.

> **Key Highlights:**
> 🚀 **Zero build step** — open `index.html` and it works
> 🤖 **Dual-path AI** — Claude API when available, comprehensive local fallback when not
> IN **Indian financial context** — INR formatting, CIBIL scoring (300–900), Indian eligibility rules
> 💎 **Premium glassmorphism UI** with 50+ CSS custom properties and micro-animations
> ☁️ **Serverless persistence** via Google Apps Script → Google Sheets

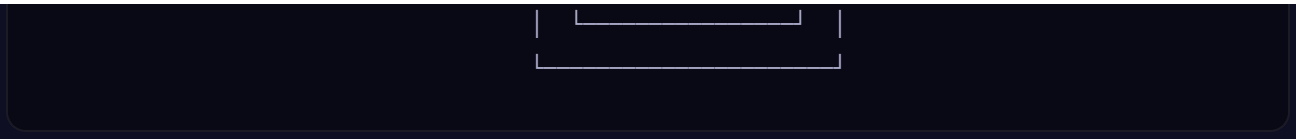
2 Project Metrics

Metric	Value
Total Files	9 (excluding `.git`)
HTML Pages	5 (`index.html`, `eligibility.html`, `credit-score.html`, `emi-calculator.html`, `ai-tips.html`)
JavaScript	1 file — `script.js` (944 lines, ~37 KB)
CSS	1 file — `styles.css` (1,382 lines, ~29 KB)
Backend Script	1 file — `google-apps-script.js` (94 lines, ~3 KB)
Total Lines of Code	~3,150+

Metric	Value
Total Project Size	~123 KB
External Dependencies	Font Awesome 6.5.1, Google Fonts (Inter, Outfit)
AI Model	Anthropic Claude (`claude-sonnet-4-20250514`)
Data Storage	Google Apps Script → Google Sheets

3 Architecture Overview





Architecture Patterns

- **Multi-page application (MPA)** — each feature is a standalone HTML page sharing one JS + CSS bundle
- **Progressive enhancement** — all modules work offline with local fallback logic; Claude AI enhances responses when configured
- **Serverless data layer** — Google Apps Script acts as a REST endpoint writing to Google Sheets, requiring zero infrastructure
- **Client-side API calls** — Claude API called directly from the browser using `anthropic-dangerous-direct-browser-access` header

4 Features Overview

📋 Loan Eligibility Checker

- Rule-based financial validation (salary, credit score, EMI, age)
- Eligible amount calculation: `Salary × 20`
- Risk classification (Low / Medium / High)
- AI-powered personalized analysis and recommendations

📊 Credit Score Analyzer

- Credit score classification (Excellent / Good / Poor)
- Animated SVG gauge visualization with gradient fill
- Actionable improvement recommendations
- AI-driven credit improvement roadmaps

🧮 EMI Calculator

- Reducing-balance EMI formula: `EMI = P × R × (1+R)^N / ((1+R)^N - 1)`
- Visual principal vs interest ratio breakdown
- Multiple loan types (Home, Personal, Car, Education, Business)
- AI-powered EMI optimization strategies

🤖 AI Financial Advisor

- Interactive chat interface powered by Claude AI
- Quick prompt suggestions for common queries
- Personalized financial guidance and planning tips
- Offline-capable with intelligent local fallback responses

☁️ Google Sheets Integration

- Serverless data storage via Google Apps Script
- Automatic record logging with timestamps

- Separate sheets for eligibility, credit, and EMI records

5 Detailed Module Breakdown

Module 1: Landing Page (`index.html` — 224 lines)

Aspect	Details
Purpose	Hero page, stats section, feature showcase, how-it-works, and CTA
Key Sections	Hero with gradient text, 4-column stats grid (99.2% accuracy, 50K+ analyses, 4.9★ rating, <2s response), 6 feature cards, 3-step how-it-works, CTA
Animations	Scroll-triggered fade-in via IntersectionObserver, floating background orbs

Module 2: Loan Eligibility Checker (`eligibility.html` — 176 lines)

Aspect	Details
Inputs	Name, Monthly Salary, Credit Score, Existing EMI, Age, Employment Type
Validation Rules	Salary ≥ ₹30,000 · Credit Score > 700 · Existing EMI < ₹20,000 · Age ≥ 21
Eligible Amount	`Salary × 20`
Risk Engine	Low (score ≥750 & EMI <10K) · Medium (score ≥650 & EMI <20K) · High (else)
AI Feature	Personalized analysis via Claude; local fallback provides structured advice
Data Logged	Name, salary, credit score, EMI, age, employment, status, eligible amount, risk level

Module 3: Credit Score Analyzer (`credit-score.html` — 190 lines)

Aspect	Details
Inputs	Credit Score (300–900), Credit History Age (years), Credit Utilization (%)

Aspect	Details
Classifications	Excellent (≥750) · Good (650–749) · Poor (<650)
Visualization	SVG semi-circle gauge with gradient fill (red→amber→green), animated via stroke-dashoffset
AI Feature	Credit improvement plan via Claude; local fallback provides category-specific roadmap
Educational	Inline score guide explaining each tier with visual cards

Module 4: EMI Calculator (`emi-calculator.html` — 204 lines)

Aspect	Details
Inputs	Loan Amount (min ₹10,000), Annual Interest Rate (0.1–36%), Tenure (1–30 yrs), Loan Type
Formula	`EMI = P × R × (1+R)^N / ((1+R)^N - 1)`
Loan Types	Home, Personal, Car/Auto, Education, Business
Visualization	Animated horizontal stacked bar showing principal vs. interest ratio
AI Feature	EMI optimization tips including prepayment savings estimates

Module 5: AI Financial Advisor (`ai-tips.html` — 176 lines)

Aspect	Details
Quick Tips	6 card-based financial tips (credit score, budgeting, EMI ratio, emergency fund, loan comparison, prepayments)
Chat Interface	Real-time conversational UI with user/AI message bubbles
Quick Prompts	4 pre-built queries (Improve Credit, Home Loan Tips, Loan Advice, Budget Rule)
API Key Mgmt	Stored in localStorage, UI auto-hides key section after save

Aspect	Details
Offline Mode	Pattern-matched local responses for credit, EMI, budget, and general queries

6 Code Architecture

JavaScript Module Map (`script.js` — 944 lines)

```
script.js
├── CONFIG                          → Google Script URL, Claude API key (localStorage)
├── Initialization (DOMContentLoaded)  → Navbar, scroll animations, forms, API key UI
├── Navigation                      → Scroll effect, mobile hamburger toggle
├── Scroll Animations               → IntersectionObserver for .animate-on-scroll
├── Form Initialization             → Event listener binding for all 3 forms
├── API Key Management              → localStorage read/write, UI toggle
├── Loan Eligibility Checker        → handleEligibilityCheck()
│   ├── AI Analysis                 → generateEligibilityAIAnalysis()
│   └── Local Fallback              → generateLocalEligibilityAnalysis()
├── Credit Score Analyzer           → handleCreditAnalysis()
│   ├── AI Analysis                 → generateCreditAIAnalysis()
│   └── Local Fallback              → generateLocalCreditAnalysis()
├── EMI Calculator                  → handleEMICalculation()
│   ├── AI Analysis                 → generateEMIAIAnalysis()
│   └── Local Fallback              → generateLocalEMIAnalysis()
├── AI Chat Interface               → sendChatMessage(), sendQuickPrompt()
│   └── Local Fallback              → generateLocalChatResponse()
├── Claude AI Integration           → callClaudeAPI() — single API wrapper
├── Google Sheets Integration       → saveToGoogleSheets() — no-cors POST
└── Utilities                       → formatNumber(), escapeHTML(), showToast()
```

Key Design Decision: Every AI-powered feature uses a dual-path architecture. If the Claude API key is configured, it calls the API; otherwise, it generates comprehensive responses using rule-based templates. This ensures 100% functionality without any API dependency.

CSS Design System (`styles.css` — 1,382 lines)

Category	Details
Fonts	Outfit (headings) + Inter (body) via Google Fonts

Category	Details
Color System	10-shade primary palette (indigo), 5 accent colors (cyan, emerald, amber, rose, violet), 10-shade dark neutral
Glass System	`rgba(255,255,255,0.05)` bg, `blur(20px)`, `rgba(255,255,255,0.1)` borders
Gradients	8 named gradient custom properties
Typography	11-step scale from 0.75rem to clamp(2.5rem, 5vw, 4.5rem)
Spacing	14-step scale (0.25rem → 6rem)
Animations	fadeInUp, float, pulse, spin, gradient text shimmer
Responsive	Mobile-first with breakpoints at 768px and 480px
Components	Navbar, glass cards, forms, result cards, gauge, chat bubbles, toasts, tips grid

Google Apps Script (`google-apps-script.js` — 94 lines)

Endpoint	Purpose
`doPost(e)`	Receives JSON payloads, routes to appropriate sheet based on `data.type`
`doGet(e)`	Health check endpoint returning `{ status: 'active' }`
`getOrCreateSheet()`	Auto-creates sheets with typed headers on first write

Auto-Created Google Sheets

Sheet Name	Columns
Eligibility Records	Timestamp, Name, Salary, Credit Score, Existing EMI, Age, Employment, Status, Eligible Amount, Risk Level
Credit Analyses	Timestamp, Score, Category, Credit Age, Credit Utilization

Sheet Name	Columns
EMI Calculations	Timestamp, Loan Amount, Interest Rate, Tenure, Loan Type, EMI, Total Interest, Total Payment

7 Technology Stack

Technology	Purpose
HTML5	Semantic structure and accessibility
CSS3	Glassmorphism design system with custom properties and animations
Vanilla JavaScript	Business logic, DOM manipulation, API integration
Claude AI (Anthropic)	Intelligent financial reasoning and personalized advice
Google Apps Script	Serverless backend for data storage
Google Sheets	Cloud-based data persistence layer
Font Awesome 6.5.1	Premium iconography
Google Fonts	Typography (Inter for body, Outfit for headings)

Eligibility Rules

Criteria	Requirement
Monthly Salary	≥ ₹30,000
Credit Score	> 700
Existing EMI	< ₹20,000
Applicant Age	≥ 21 years
Eligible Amount	**Salary × 20**

8 Security Considerations

Area	Status	Notes
XSS Protection	✅ Implemented	`escapeHTML()` sanitizes all user input before DOM insertion
API Key Storage	⚠️ Client-side	Claude API key stored in localStorage — acceptable for personal use
CORS	⚠️ Browser-direct	Uses `anthropic-dangerous-direct-browser-access` header — dev/personal use
Data Transmission	✅ Secure	Google Sheets POST uses `mode: 'no-cors'`
Input Validation	✅ Implemented	All form inputs validated with range checks before processing

9 Performance Analysis

Aspect	Details
Bundle Size	~123 KB total (no frameworks, no build step)
Load Time	Instant — static HTML, single JS + CSS file
Runtime	All computation is client-side; API calls are async
Animations	Hardware-accelerated CSS transforms, IntersectionObserver for lazy triggering
Number Formatting	Indian numbering system (lakhs/crores) via custom `formatNumber()`

10 Future Improvements

Area	Suggestion
Amortization Schedule	Add a monthly breakdown table to the EMI calculator
Charts	Integrate Chart.js or D3.js for pie/doughnut charts in EMI breakdown
PWA Support	Add manifest.json and service worker for installable offline support
Cross-page State	Use sessionStorage or URL params to share data between modules
Testing	Add unit tests for formatNumber(), EMI calculation, and eligibility logic
Accessibility	Add ARIA labels to dynamic result sections, improve keyboard navigation
API Rate Limiting	Add client-side rate limiting for Claude API calls

Area	Suggestion
Multi-language	Add Hindi and regional language support for broader Indian audience